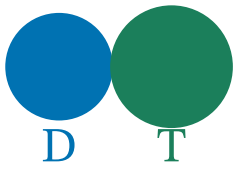
Power balance and Lawson

Christopher Albert

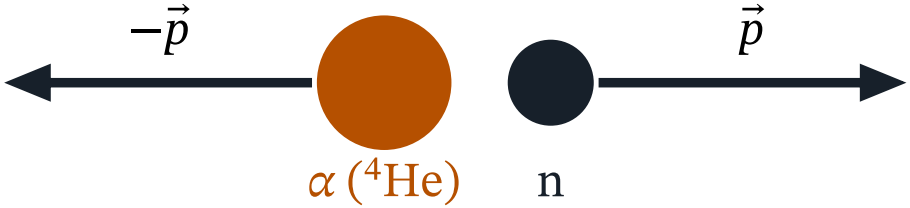
4.1 Alpha and neutron

before



$p_{\text{tot}} \approx 0$

after (centre-of-mass frame)



$$E = \frac{p^2}{2m} \Rightarrow \frac{E_\alpha}{E_n} = \frac{m_n}{m_\alpha} \approx \frac{1}{4}$$

alpha (20 %)

$Q_{\text{DT}} = 17.59$ MeV



with relativistic kinematics: 3.56 MeV and 14.03 MeV

$$E_\alpha = 3.5 \text{ MeV}, \quad E_n = 14.1 \text{ MeV}$$

α heats the plasma, n the blanket

4.2 Fusion power density

fuel $n_\mathrm{D} + n_\mathrm{T} = n, \quad n_\mathrm{D} n_\mathrm{T} = n^2 x(1-x)$

optimum $x = 1/2$

$$p_\alpha = \frac{n^2}{4}\langle\sigma v\rangle E_\alpha, \quad p_\mathrm{fus} = \frac{n^2}{4}\langle\sigma v\rangle E_\mathrm{fus}$$

$n = 10^{20}\mathrm{m}^{-3}$, 14 keV: $p_\mathrm{fus} = 1.7\ \mathrm{MW/m^3}$

4.3 Stored energy and confinement time

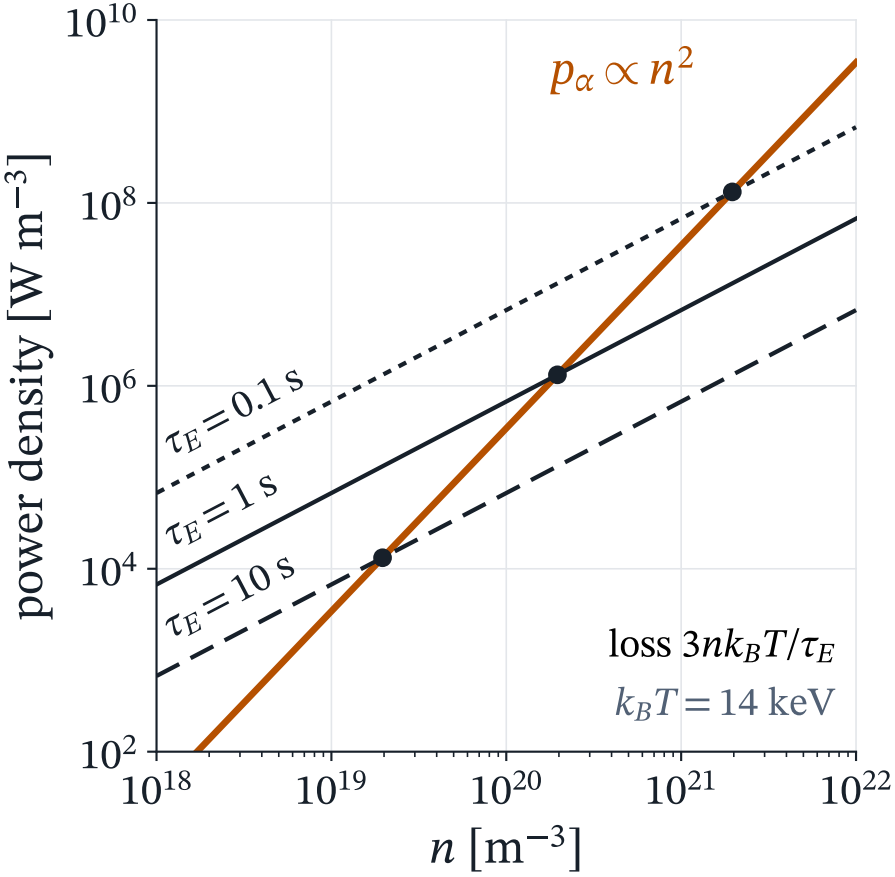
quasineutral $n_e = n_\mathrm{D} + n_\mathrm{T} = n$

stored $W = 3nk_BT$

losses $p_\mathrm{loss} = W/\tau_E$

$$p_\mathrm{ext} + p_\alpha = W/\tau_E$$

steady state

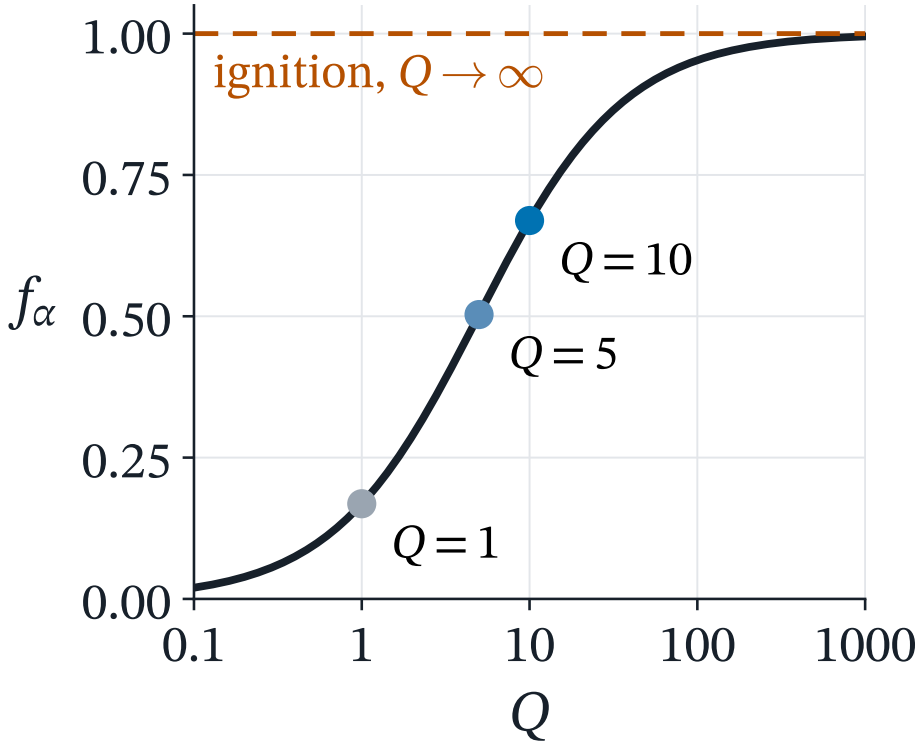


4.4 Power balance with gain

gain $Q = p_{\text{fus}}/p_{\text{ext}}$

$E_{\text{fus}} \approx 5E_\alpha$ $f_\alpha = \dfrac{p_\alpha}{p_\alpha + p_{\text{ext}}} = \dfrac{Q}{Q+5}$

Q		f_α
1	break-even	1/6
5	burning plasma	1/2
∞	ignition	1



4.4 Triple product

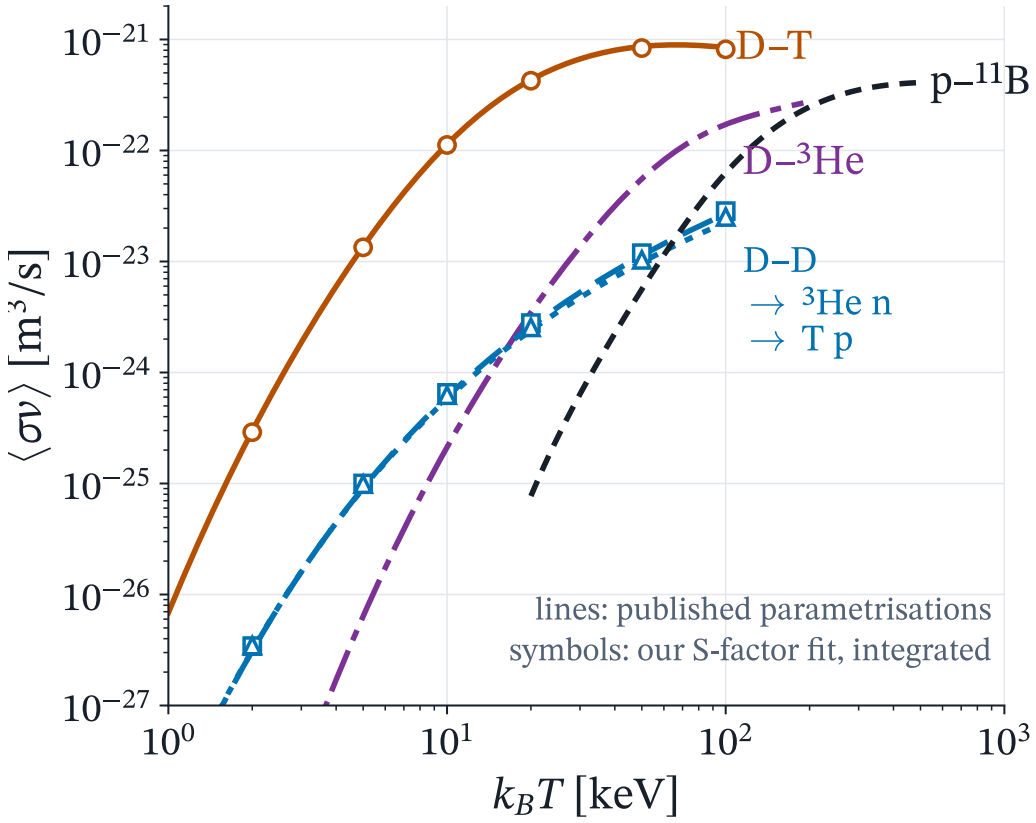
ignition $\frac{n^2}{4}\langle\sigma v\rangle E_\alpha = \frac{3nk_BT}{\tau_E}$

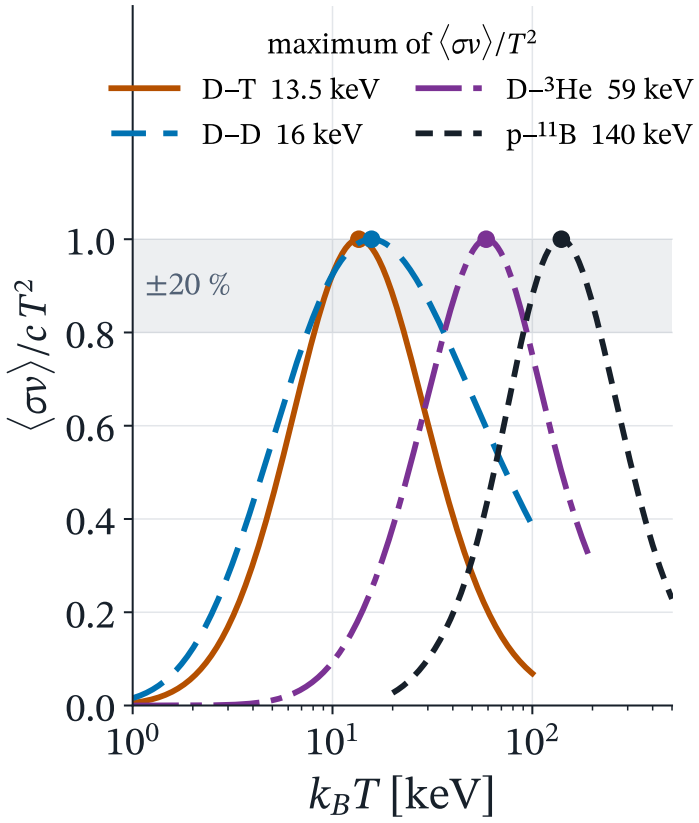
10–20 keV $\langle\sigma v\rangle \propto T^2$ so $nT\tau_E$ is constant

$$nk_BT\tau_E = \frac{12(k_BT)^2}{\langle\sigma v\rangle E_\alpha} \approx 3\times10^{21}\,\mathrm{keV\,s\,m^{-3}}$$

= 5 bar s ion pressure, 10 bar s ion + electron pressure

4.4 How good is $\langle \sigma v \rangle \approx c T^2$?



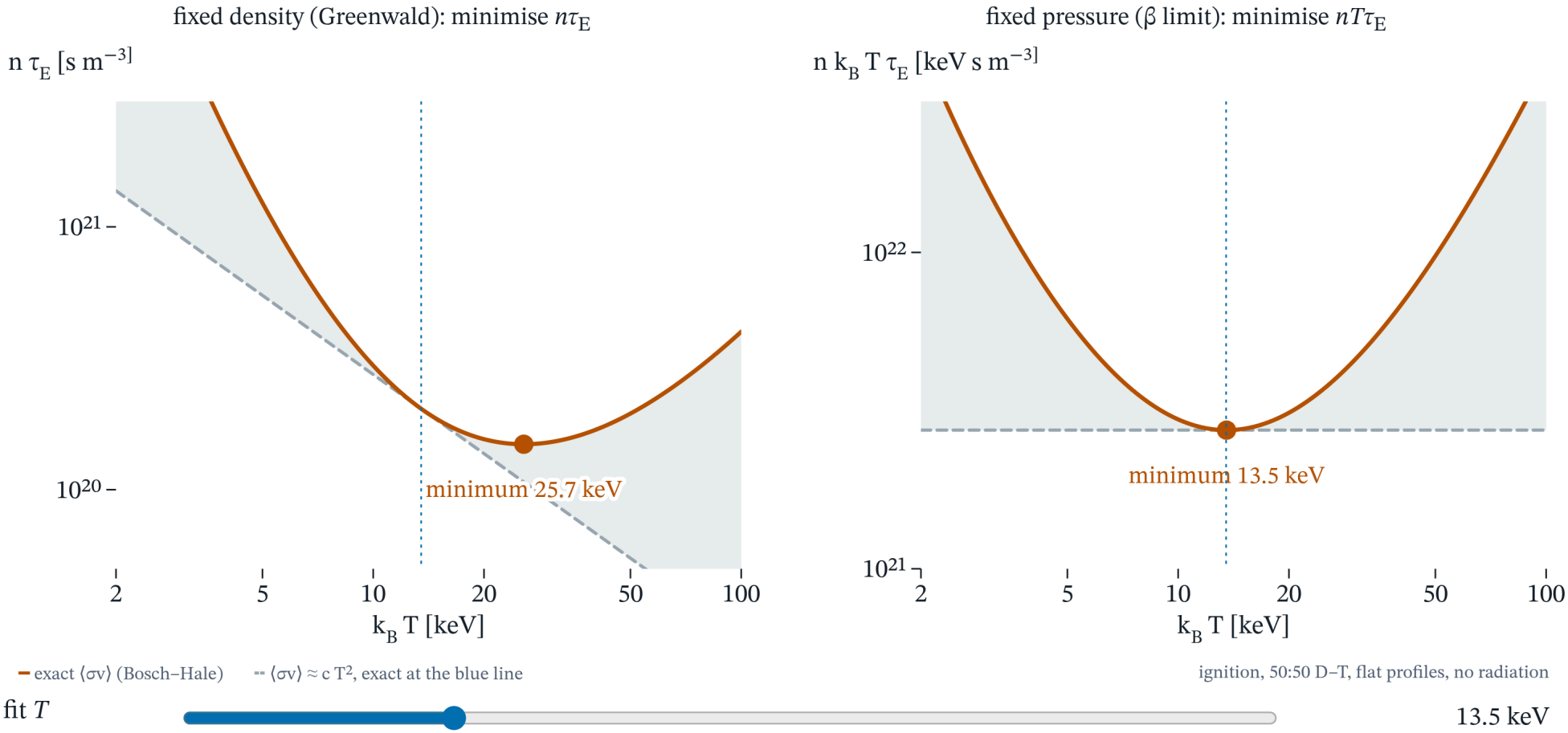


within 20 %: **D–T 8–23 keV** D–D 7.5–36 keV D–^{3}He 36–94 keV p–^{11}B 91–218 keV

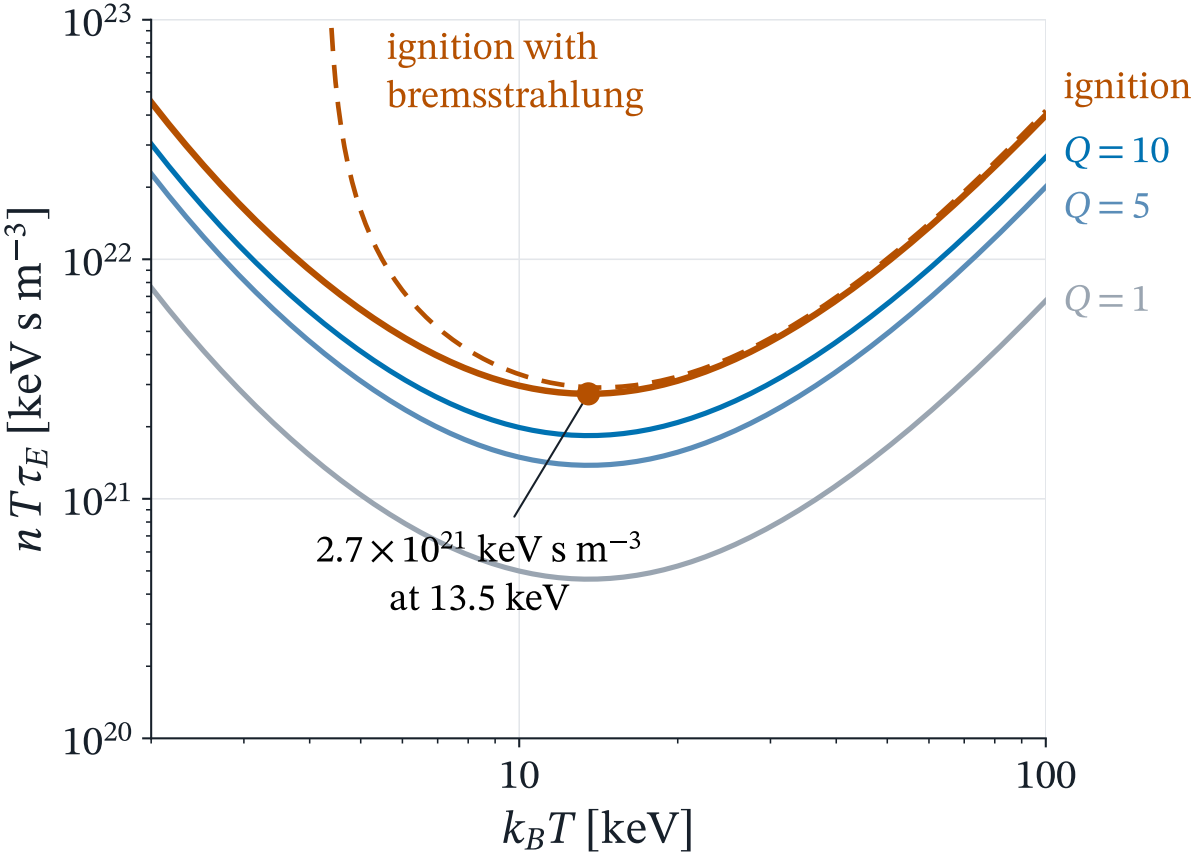
c: maximum of $\langle \sigma v \rangle / T^2$ for each fuel (grey band)

Bosch & Hale · Nucl. Fusion 32, 611 (1992) · Nevins & Swain · Nucl. Fusion 40, 865 (2000)

4.4 $n\tau_E$ or $nT\tau_E$?



4.4 Lawson plane



gain Q $\quad nT\tau_E \propto \dfrac{Q}{Q+5}$

$Q = 10$: 2/3

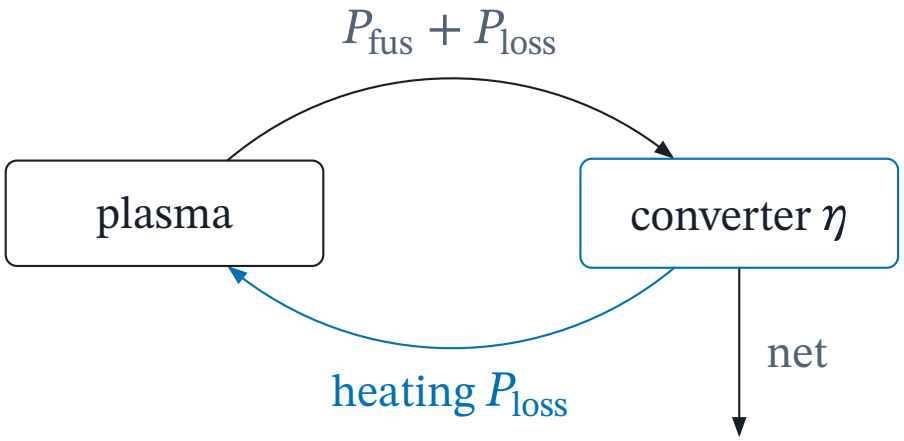
of the ignition value;
minimum near 13.5 keV

dashed: with bremsstrahlung (4.7)

Bosch & Hale · Nucl. Fusion 32, 611 (1992) · notebook: Lawson criterion

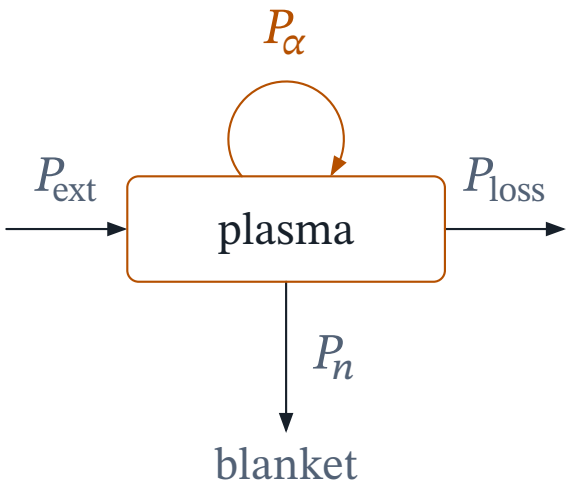
4.5 Lawson 1957 and self-heating

Lawson 1957



$$P_{\text{fus}} \geq 2P_{\text{loss}} \quad (\eta = 1/3)$$

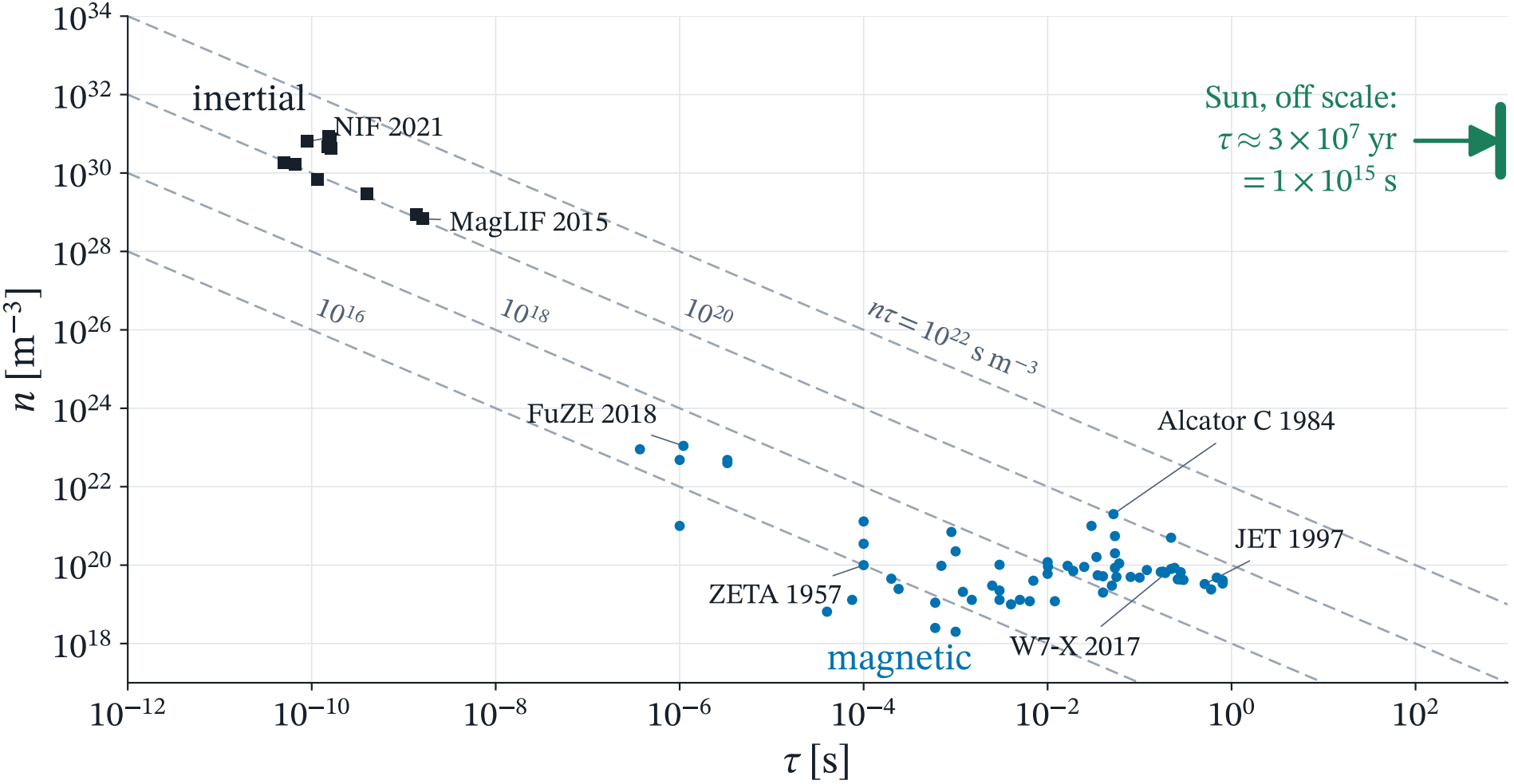
Self-heating



$$P_\alpha \geq P_{\text{loss}} \quad \text{(ignition)}$$

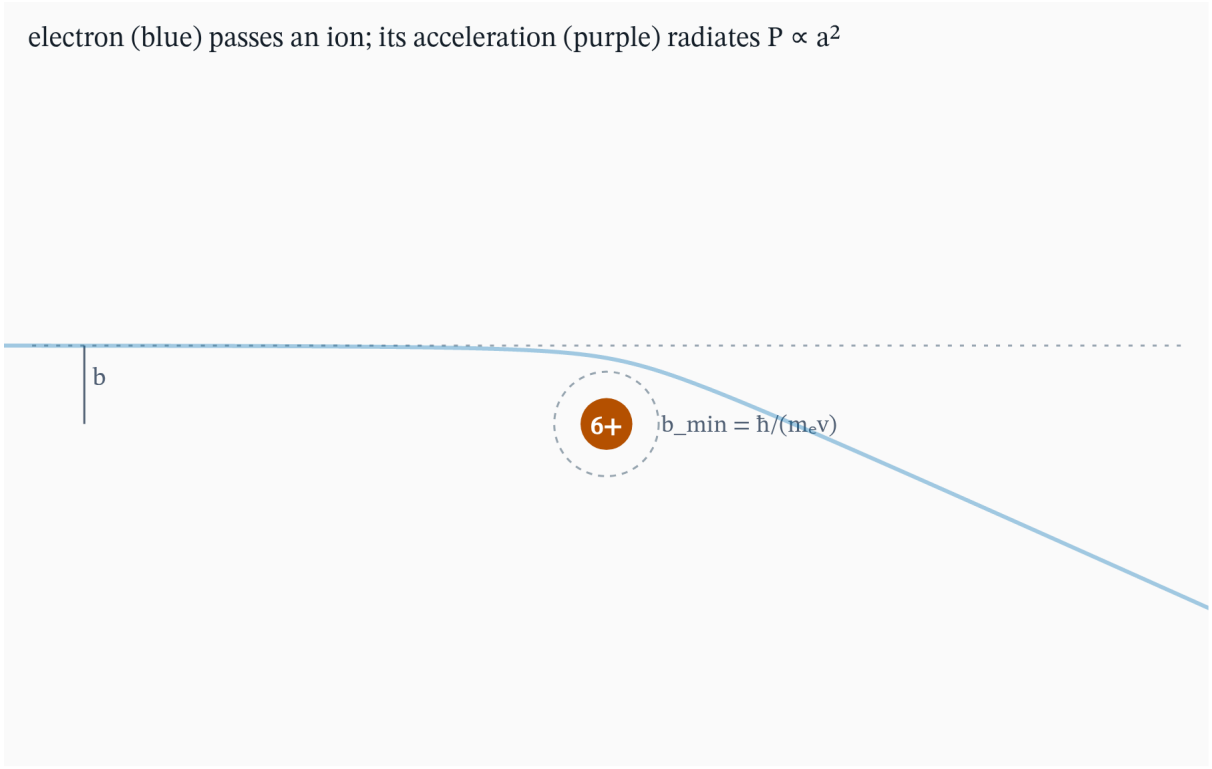
Lawson · Proc. Phys. Soc. B 70, 6 (1957) · Wurzel & Hsu · Phys. Plasmas 29, 062103 (2022)

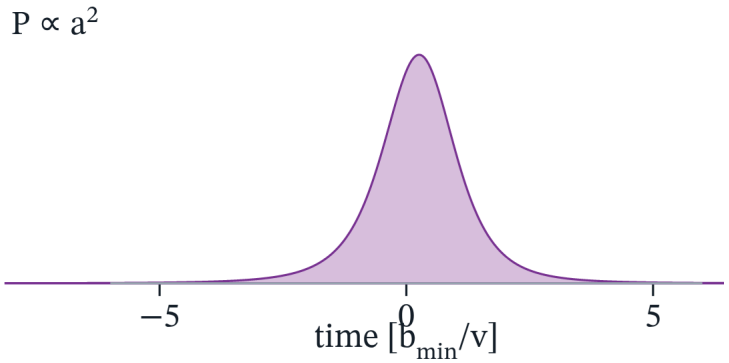
4.6 Density and confinement time

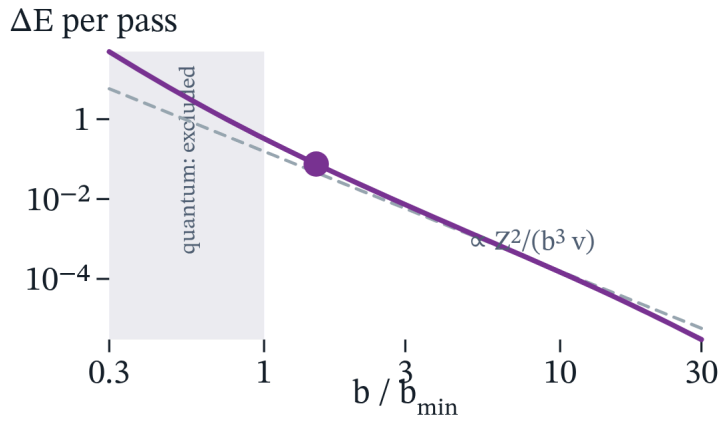


Data: Wurzel & Hsu · Phys. Plasmas 29, 062103 (2022), Tables 6–8, CC BY 4.0

4.7 How bremsstrahlung arises







$\kappa = Z\alpha c/v = 0.315 \cdot v = 0.139\ c$

$\kappa < 1$: the path bends noticeably; far passes radiate $\Delta E \propto Z^2/(b^3 v)$

this pass: 1.63 × the straight-line value

b / b_{min} 1.50

E_e [keV] 5.0 keV

ion Z ○ 1 ○ 2 ○ 4 ◉ 6 ○ 8

4.7 Bremsstrahlung from one pass

Larmor $$P = \frac{e^2 a^2}{6\pi\varepsilon_0 c^3}$$

Coulomb kick $$a \approx \frac{Ze^2}{4\pi\varepsilon_0 m_e b^2}, \quad \Delta t \approx \frac{2b}{v}$$

one pass $$\Delta E = P\Delta t \propto \frac{Z^2}{b^3 v}$$

passes $n_i v\, 2\pi b\, db$ per second, $b \geq b_{\min} = \hbar/(m_e v)$

one electron $$P_e = \int_{b_{\min}}^{\infty} \Delta E\, n_i v\, 2\pi b\, db \propto Z^2 \frac{n_i}{b_{\min}} \propto Z^2 n_i v$$

$$p_{\mathrm{br}} = C_B\, Z^2 n_e n_i \sqrt{k_B T}$$

Maxwell: $\langle v \rangle \propto \sqrt{T}$ estimate 8.4, exact $5.34 \times 10^{-37}\,\mathrm{W\,m^3 keV^{-1/2}}$

NRL Plasma Formulary · Jackson, Classical Electrodynamics, §15.2

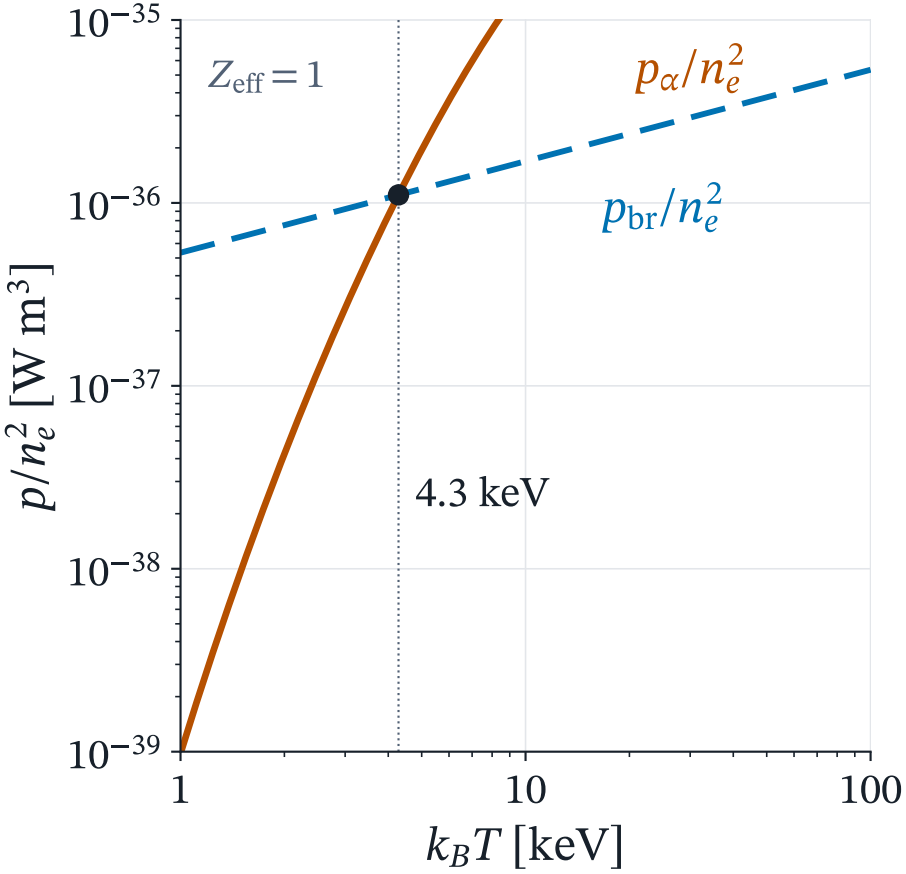
4.7 Bremsstrahlung and ideal ignition

electron–ion $p_{\mathrm{br}} = C_B n_e^2 Z_{\mathrm{eff}} \sqrt{k_B T}$

$p_\alpha = p_{\mathrm{br}}$ $\langle \sigma v \rangle E_\alpha = 4 C_B Z_{\mathrm{eff}} \sqrt{k_B T}\ n$ cancels

$$k_B T_{\mathrm{ideal}} = 4.3\ \mathrm{keV}$$

D–T, $Z_{\mathrm{eff}} = 1$, any density

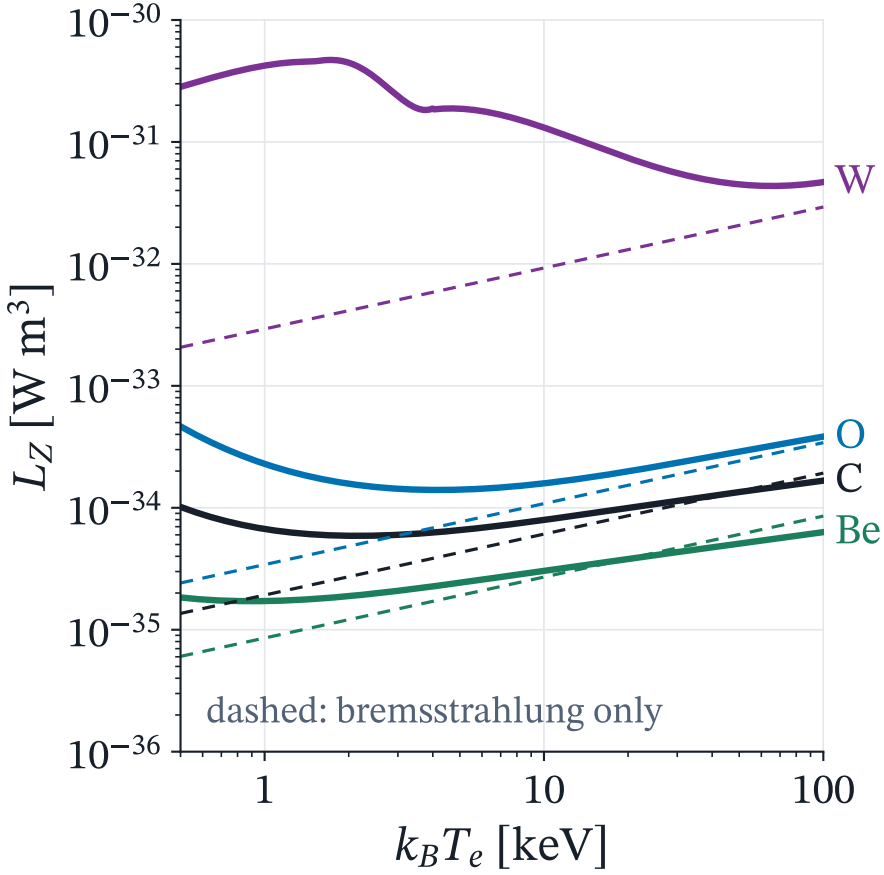


$C_B = 5.34 \times 10^{-37}\ \mathrm{W\ m^3 keV^{-1/2}}$

Lawson · Proc. Phys. Soc. B 70, 6 (1957) · NRL Plasma Formulary

4.7 Line radiation

excite	electron hits ion, bound electron jumps up
decay	photon $h\nu = E_{\text{upper}} - E_{\text{lower}}$
coronal	every excitation ends in a photon
power	$p_Z = n_e n_Z L_Z(T_e)$

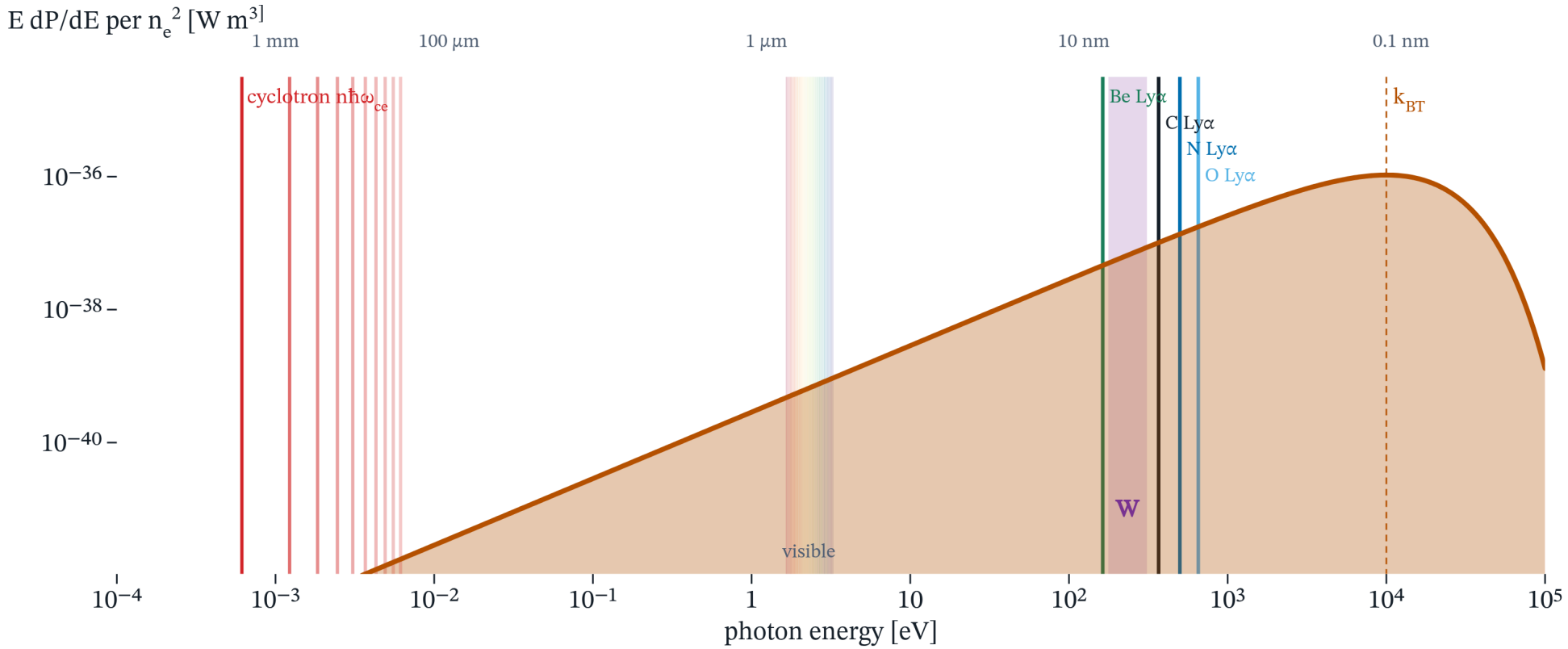


needs bound electrons

light ions bare at 10 keV: $L_Z \to Z^2 C_B \sqrt{T}$
W keeps electrons: $L_W \approx 10^{-31}$W m^3

Mavrin · Radiat. Eff. Defects Solids 173, 388 (2018) · Pütterich et al. · Nucl. Fusion 50, 025012 (2010)

4.7 Where a plasma radiates



bremsstrahlung: continuum up to a few $k_B T$, 90 % of its power above 1 keV (X-rays) — leaves the plasma

cyclotron at 0.61 meV × n (mm waves) — reabsorbed

lines: bound electrons of impurities

Kramers spectrum, Gaunt factor 1; lines and harmonics mark positions only (H-like Lyα from the Bohr model; W quasi-continuum 4–7 nm: Pütterich et al., Plasma Phys. Control. Fusion 50, 085016, 2008).

$k_B T_e$ — 10.0 keV

B — 5.3 T

Z_{eff} — 1.7

4.7 Cyclotron radiation

one electron $$P = \frac{e^4 B^2 v_\perp^2}{6\pi\varepsilon_0 m_e^2 c^3}$$

Maxwellian $$p_{\mathrm{cyc}} \approx 6.2 \times 10^{-17}\, n_e\, k_B T_e\, B^2 \ \mathrm{W\,m^{-3}}$$

$10^{20}\,\mathrm{m^{-3}}$, 10 keV, 5 T [MW m^{-3}]:

p_{fus}	p_{br}	p_{cyc} emitted
0.8	0.017	1.6

optically thick: mostly reabsorbed, walls reflect

grows with T_e and B: a main loss for D–^{3}He and p–^{11}B

n_e in m^{-3}, $k_B T_e$ in keV, B in T pairs need $2m_e c^2 = 1.02$ MeV: negligible for $k_B T \ll 511$ keV

NRL Plasma Formulary

4.8 Quasineutrality and dilution

quasineutral $n_e = n_\mathrm{D} + n_\mathrm{T} + \sum_Z Z n_Z \quad c_Z = n_Z/n_e$

fuel $f = (n_\mathrm{D} + n_\mathrm{T})/n_e = 1 - \sum_Z Z c_Z, \quad p_\mathrm{fus} \propto f^2 n_e^2$

stored $W = \frac{3}{2}(n_e + n_i) k_B T$

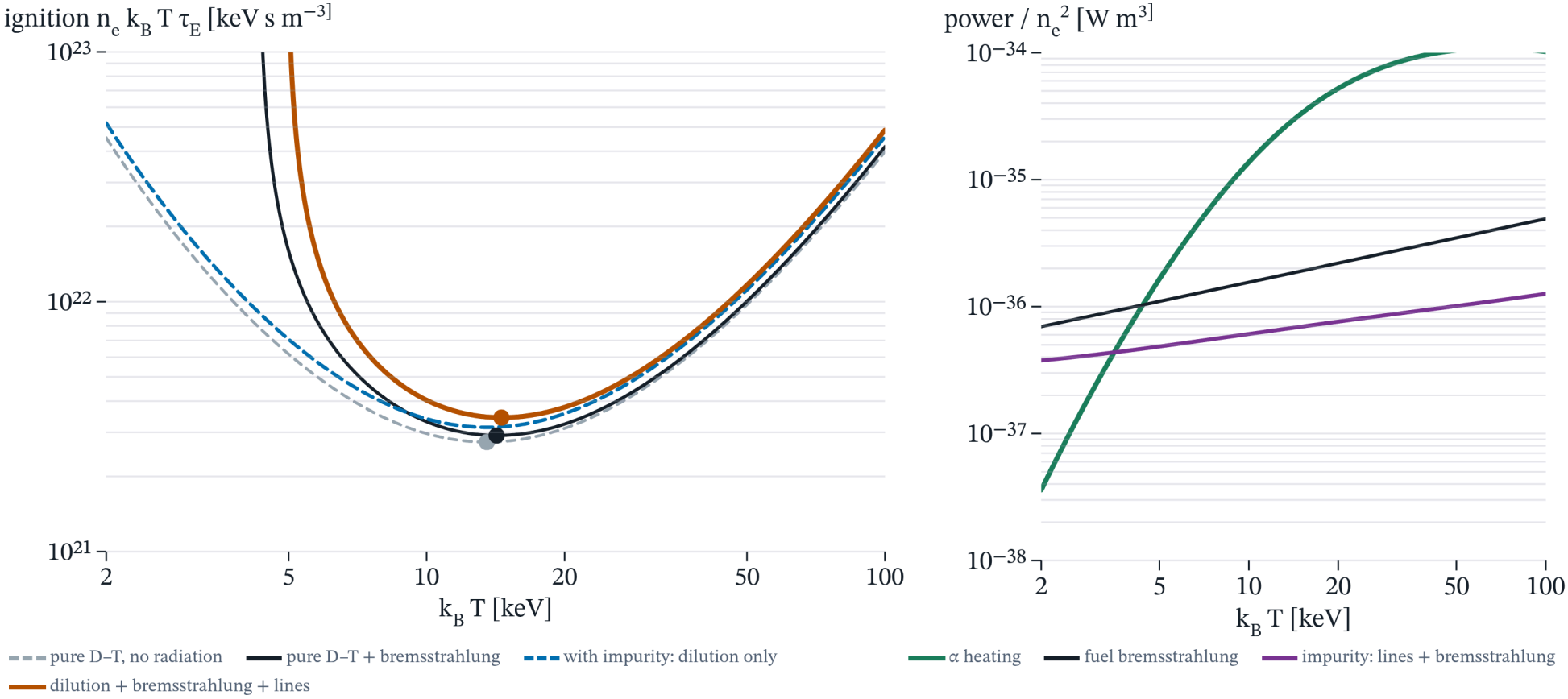
losses $C_B f n_e^2 \sqrt{T}\,\text{(fuel)} + \sum_Z n_e n_Z L_Z(T)\,\text{(impurities)}$

$$n_e T \tau_E = \frac{\frac{3}{2}(1 + n_i/n_e) T^2}{f^2 \langle \sigma v \rangle E_\alpha / 4 - C_B f \sqrt{T} - \sum_Z c_Z L_Z(T)}$$

W: displaces 74 fuel ions and radiates; $c_W = 3 \times 10^{-5}$: minimum +9 % (0-D, coronal)

Pütterich et al. · Nucl. Fusion 50, 025012 (2010) · Mavrin · Radiat. Eff. Defects Solids 173, 388 (2018)

4.8 Impurities: dilution and radiation



fuel $n_D + n_T = 92.0\ \%$ of $n_e \cdot p_{fus} \times 0.847 \cdot Z_{eff} = 1.24$ · minimum 3.44×10^{21} at 14.6 keV, ×1.25 of pure D–T

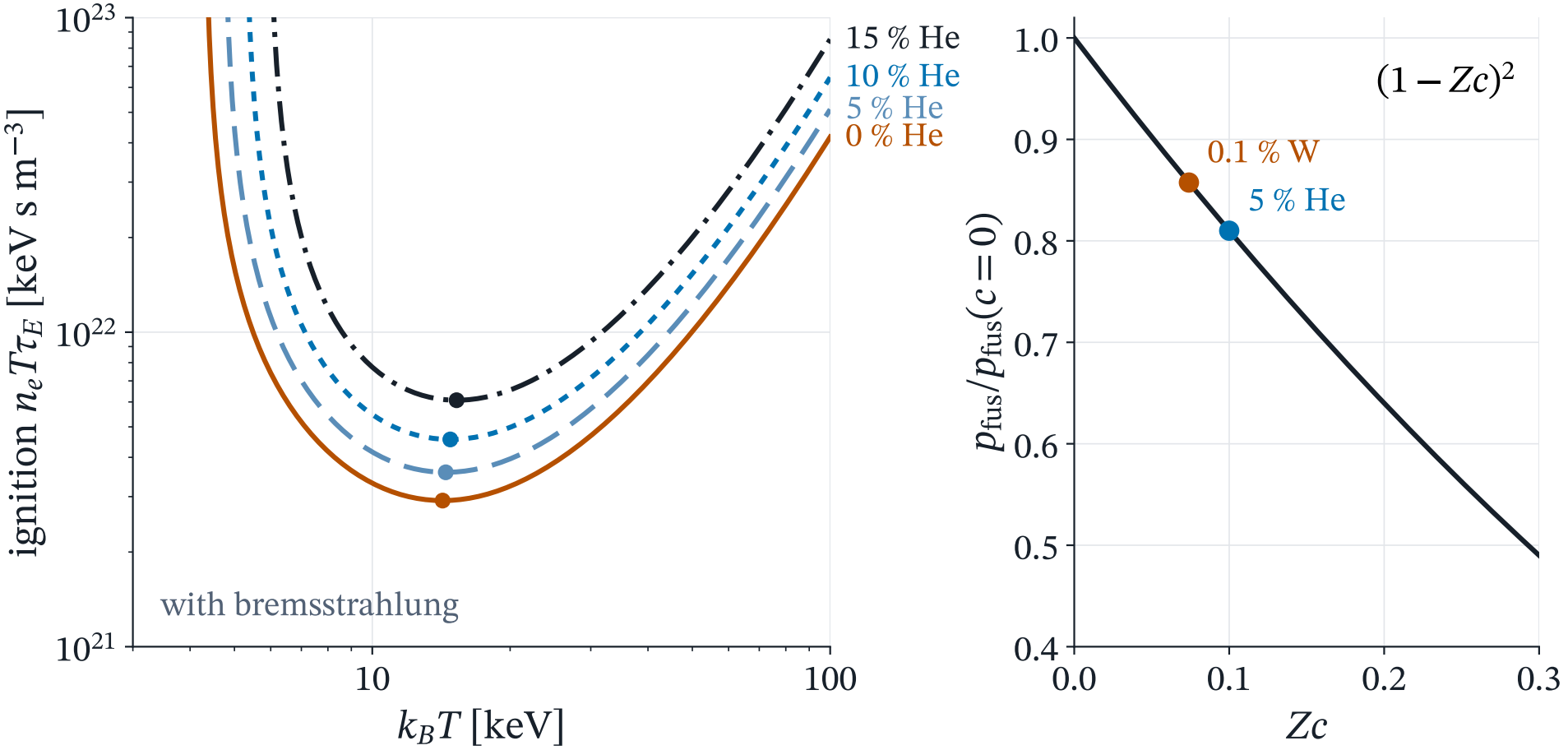
impurity: Be (Z = 4)

n_Z/n_e: 2.0 %

bremsstrahlung ☑

line radiation ☑

4.8 Helium ash



helium concentration c: fuel $1-2c$, $Z_{\mathrm{eff}} = 1+2c$

notebook: impurities and fuel dilution